

Utilizing Professional Development to Improve Student Career Outcomes in Computer Science & Software Engineering

Clint Lovelace
CSSE Department
Auburn University
Auburn, AL United States
jcl0014@auburn.edu

Michael Covington
CSSE Department
Auburn University
Auburn, AL United States
jmc0031@auburn.edu

Matthew Morris
CSSE Department
Auburn University
Auburn, AL United States

ABSTRACT

This project has two objectives: (1) collect 100% detailed first destination data from all students graduating from the CSSE department (between the summer 2020 and spring 2021 semesters, only 83% of students completed the First Destination Survey), and (2) enhance first career outcomes for women and African-American graduates (of those who completed the survey, 39% of women and 69% of African Americans responded with “still seeking employment/continuous education”). Accomplishing a higher response rate and drilling down into the data to better understand why women and minorities are still seeking opportunities at graduation time will allow for more strategic and effective interventions to improve CSSE student career outcomes. These two objectives will be met through a professional development sequence, a program assessment course, and the creation of a professional mentoring program for seniors with working professionals and a peer mentoring program for sophomores mentored by juniors. The professional development sequence includes topics such as effective resumes and cover letters, professional communication, interviewing, networking, and continuing education. The two courses will include additional data collection to better understand and track first destination success for women, first generation students, low SES (e.g. Pell grant) students, and African-American students as well as to assess the success of these interventions in enhancing first destination success for all CSSE students.

Professional Development Sequence

CSSE has proposed a professional development sequence that will directly align with the QEP objectives, both in terms of quantity and quality of student placement: COMP 2800, COMP 4800, and COMP 4810. Additionally, students enrolled in the seminar sequence will participate in a peer and professional mentoring program. First Destination data will be collected through the required COMP 4810 Program Assessment course. Through these three classes, CSSE aims to improve completion percentage of the

First Destination Survey, as well as improve both the quantity and quality of student first destination success within the Department of Computer Science and Software Engineering.

COMP 2800 Professional Development I will be a 0-credit hour required course. The course aims to provide students a more complete understanding of the research subareas and career paths in the field of computer science and software engineering, develop strategies to leverage personality strengths for academic and career success utilizing the Clifton Strengths Assessment, develop a high-quality resume, learn the importance of professional communication, explore the various industries that hire CSSE majors and develop strategies to increase career readiness, and understanding using appropriate learning strategies to acquire and apply new knowledge.

COMP 4800 Professional Development II is a 0-credit hour course that aims to prepare students to communicate effectively in a variety of professional contexts, develop interview skills to improve career readiness, develop strategies to utilize a digital presence for career success, learn to function effectively as a member or leader of a team, and acquire and apply new knowledge as needed throughout their career. Throughout COMP 4800, students will explore industry, develop and refine resumes and cover letters, improve interview skills, build a personal and professional brand, understand how to navigate job offers, and learn the importance of effective networking.

COMP 4810 Program Assessment will utilize required surveys to collect program, departmental, and First Destination data. Surveys will be given to gather qualitative and quantitative data on the overall student experience, including curriculum, course offerings, student services, and the overall program experience, aligning with the Accreditation Board for Engineering and Technology standards.

REFERENCES

- [1] Buckley-Burnell, A. E. (2021). Examining the Relationships Between Mandatory Career Courses, High-Impact Practices, and First-Destination Outcomes (dissertation).

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